

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027

Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer on a provider-facing clinical decision support app (Next.js + Firebase/GCP) that surfaces covered, guideline-backed medication alternatives in the EHR prescribing flow, work aimed directly at improving patient care.
- Extending a backend (Firebase Cloud Functions) that grounds recommendations in **RxNorm** normalization and PubMed/clinical-guideline evidence, with **Gemini** constrained to synthesis while the backend deterministically controls results, keeping clinical AI output evidence-based and verifiable.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building a research ML platform (Python, **FastAPI**, **sentence-transformers**, React) for UMass's Culture and Morality Lab, productionizing a published embedding-based text-measurement method into a self-serve tool for researchers.
- Built a YAML model registry, a versioned construct library (**99 constructs from 38 questionnaires**), and per-run reproducibility exports with structured data-quality checks; grew the backend to **40 hermetic tests** in CI, prioritizing reproducible, rigorously validated analysis.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Architected a full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by the AI team, replacing spreadsheet workflows and cutting manual annotation by **100+ hours/month** while lifting labeling throughput **60%**.
- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly submissions** via REST integrations; contributed to **\$340K+** in revenue.

Projects

align-ops – LLM Post-Training + Serving Ops PyTorch, TRL (SFT/DPO/GRPO), vLLM, FastAPI, Docker

- Built an end-to-end LLM post-training pipeline (PyTorch, TRL) through supervised fine-tuning, DPO, and GRPO, lifting held-out task accuracy **54% to 72%**, and served the model via a vLLM + FastAPI gateway with Redis caching and async micro-batching for **1.9x throughput**, instrumented with Prometheus/Grafana.

Refusal Decay – Mechanistic Interpretability / LLM Safety (GitHub) PyTorch, Llama-3.1-8B, AdvBench

- Ran a rigorous mechanistic-interpretability study (PyTorch) of how prefilling attacks weaken refusal in Llama-3.1-8B-Instruct, extracting the residual-stream refusal direction via difference-in-means with causal patching, bootstrap 95% CIs, and McNemar's test.

KG2RAG-Enhanced – Multi-Hop QA on HotpotQA (GitHub) Python, Ollama / LLaMA-3, sentence-transformers, RRF

- Built the multi-view retrieval component of a KG-guided RAG pipeline (sentence-transformers, RRF fusion, cross-encoder reranking, MMR), lifting supporting-fact recall **+2.68%** over the KG²RAG baseline on a 4,905-question evaluation with 95% binomial CIs.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: Python, C++, SQL, JavaScript, TypeScript, Bash

ML & DL: PyTorch, TensorFlow, JAX, Hugging Face Transformers, sentence-transformers, generative models, NumPy

Computer Vision: CNNs (VGG/ResNet), transfer learning, medical image classification, MediaPipe, model evaluation

MLOps & Serving: TRL (SFT/DPO/GRPO), QLoRA/PEFT, vLLM, FastAPI, Prometheus/Grafana, Spark/PySpark

Web & Cloud: React, Next.js, Node.js, AWS, GCP, Azure, Docker, Git

Open Source & Community

- Built **Underdogs Fitness** (Website), a production MERN platform with Stripe payments; live **3+ years**. - Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.